TAYLOR'S UNIVERSITY
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TAYLOR'S UWE DUAL AWARDS PROGRAMMES

SEPTEMBER 2026 SEMESTER

DEEP LEARNING

(CSC60904/CSC62904)

Group Assignment (30%)

DUE DATE: 07th December 2026 via MyTIMeS (11:59 pm)



STUDENT DECLARATION

1. *I confirm that I am aware of the University's Regulation Governing Cheating in a University Test and Assignment and of the guidance issued by the School of Computing and IT concerning plagiarism and proper academic practice and that the assessed work now submitted is by this regulation and guidance.*
2. *I understand that, unless already agreed with the School of Computing and IT, assessed work may not be submitted that has previously been submitted, either in whole or in part, at this or any other institution.*
3. *I recognise that should evidence emerge that my work fails to comply with either of the above declarations, then I may be liable to proceedings under Regulation.*

Student Name	Student ID		Signature	Score

SEPTEMBER 2026 SEMESTER GROUP ASSIGNMENT	30%

Table of Contributions:

Group no: …

Group leader: …

No.	Student Name	Student ID	List of Tasks/Works Done	Contribution Percentage
1				
2				
3				
4				
5				
6				

SEPTEMBER 2026 SEMESTER GROUP ASSIGNMENT	30%

GROUP PROJECT

A COMPARATIVE ANALYSIS OF GRADIENT DYNAMICS AND INFORMATION FLOW: EVALUATING BACKPROPAGATION STABILITY IN DEEP RESIDUAL AND INCEPTION ARCHITECTURES

Total Marks: 100

Instruction for Students: Think of this report as a Review Article. Your goal is to explain to the reader why deep learning architectures evolved from simple stacks (VGG) to complex topological graphs (ResNet/Inception) to solve the fundamental physics of the Backpropagation algorithm.

Note: Each group must select **ONE** of the following domain options to complete Part 1 and Part 2 of the assignment.

Option	Domain	Source of Dataset		
A	Medical Imaging ISIC Skin Lesion Classification	**Kaggle ISIC 2018 Dataset:** ISIC 2018 Skin Lesion on Kaggle	**Kaggle ISIC 2019 Dataset:** ISIC 2019 Skin Lesion on Kaggle	**Kaggle SIIM-ISIC Competition:** SIIM-ISIC Melanoma Classification
B	Remote Sensing EuroSAT Satellite Land Use Classification	**Official GitHub Repository:** phelber/eurosat on GitHub	**Kaggle Dataset:** EuroSAT Dataset on Kaggle	**TensorFlow Datasets API:** Available directly via tfds.load('eurosat/rgb') or tfds.load('eurosat/all')
C	Acoustic Signals UrbanSound8K Spectrogram Classification	**Official Website:** UrbanSound8K Official Portal	**Kaggle Dataset:** UrbanSound8K on Kaggle	
D	Agriculture PlantVillage Leaf Disease Classification	**Official GitHub Repository:** spMohanty/PlantVillage-Dataset on GitHub	**Meta-Album Dataset Portal:** PlantVillage on Meta-Album	**TensorFlow Datasets API:** Available directly via tfds.load('plant_village')
E	Autonomous Driving GTSRB (German Traffic Sign Recognition Benchmark)	**Kaggle Dataset:** GTSRB German Traffic Sign Dataset on Kaggle	**Torchvision / TF Datasets API:** Available via torchvision.datasets.GTSRB	
F	Retinal Ophthalmology APTOS 2019 Blindness Detection	**Kaggle Competition & Dataset:** APTOS 2019 Blindness Detection on Kaggle		
G	Food Visual Categorization Food-101 Dataset	**Official ETH Zurich Portal:** ETH Zurich Food-101 Web Portal	**Kaggle Dataset:** Food-101 on Kaggle	**TensorFlow Datasets API:** Available directly via tfds.load('food101')
7	Industrial Inspection NEU Surface Defect Database	**Kaggle Dataset:** NEU Surface Defect Dataset on Kaggle		

Part 1: Research and Critical Analysis of Loss and Gradient Flow

Total:(40 Marks)

Question 1. Research and critically evaluate the backpropagation mechanism within three supervised architectures (e.g., VGGNet, ResNet, and Inception). Your analysis must be framed as a comparative study of optimization stability.

1.1. Explain how stacking standard layers in VGGNet causes the Vanishing Gradient Problem as network depth increases. Refer to the Chain Rule (multiplying partial derivatives layer-by-layer) in your answer. Explain how weight values and activation functions (e.g., ReLU) affect gradient stability in the earliest layers. Include a simple graph showing gradient magnitudes across layers for VGG versus ResNet.

(10 Marks)

1.2. Use the residual function $F(x) = H(x) - x$ to explain how "shortcut connections" modify the gradient flow. Explain why ResNet can build much deeper networks (e.g., 152 layers) without increasing training error, whereas plain non-residual networks see an increase in error as depth grows.

(10 Marks)

1.3. The Inception (GoogLeNet) architecture utilizes "Inception Modules" to process information in parallel. Explain how the use of $1x1, 3x3, and\ 5x5$ Filters in parallel affect the model's ability to handle objects of varying scales within an image. Explain why 1×1 Convolutions are used to reduce channel depth before applying larger filters, keeping computational costs manageable during backpropagation.

(10 Marks)

1.4. Explain why pairing Softmax activation with Categorical Cross-Entropy loss is better for backpropagation in classification than using Mean Squared Error (MSE), specifically regarding neuron derivative saturation. Choose **one model** (VGG, ResNet, or Inception) as your backbone for Part 2 and provide a clear technical defense justifying why it is the best fit for your chosen dataset.

(10 Marks)

Part 2: Implementation and Evaluation

Question 2. Implement the chosen classification model (selected in Part 1) to perform image recognition on a public dataset (or a simple custom dataset). **Total: (60 Marks)**

- **Implementation Steps (to be included in your report and comments in Jupyter Notebook):**

2.1. List the deep learning libraries used to build your network.

2.2. Explain how your framework tracks operations during the Forward Pass to calculate derivatives automatically in the Backward Pass (e.g., PyTorch's requires_grad=True or TensorFlow's tf.GradientTape).

2.3. Explain the memory difference between training mode and evaluation mode (e.g., torch.no_grad()). **(5 Marks)**

2.4. Implement data preprocessing, augmentation, and label encoding for your chosen dataset option.

2.5. Pass a small batch of images through your data pipeline. Apply two random augmentations (e.g., rotation, color jitter) to a single sample and display the original vs. augmented version side-by-side.

2.6. Explain how data augmentations increase dataset variance without changing class labels.

(15 Marks)

2.7. Build your chosen model (selected in Part 1) and print the .summary() output, showing layer shapes, types, and total trainable parameter counts.

2.8. Pass a test sample through the network, extract feature maps from one intermediate hidden layer, and display a small grid of these active feature maps.

2.9. Explain why spatial dimensions contract (height/width decrease) while channel depth expands as data moves deeper into the network, and how backpropagation uses trainable parameters to update weights. **(15 Marks)**

2.10. Train your model using an adaptive optimizer (e.g., Adam or RMSprop) over a fixed number of epochs.

2.11. Plot training and validation loss curves on a dual-axis graph, overlaid with the learning rate trajectory or gradient norms across epochs.

2.12. Explain the basic principles of your chosen optimizer and how it uses past gradient information (moments) to adjust updates.

2.13. Analyze your loss curves: identify patterns such as plateaus, underfitting, or the exact epoch where overfitting begins. **(10 Marks)**

2.14. Evaluate your model on the test set and print the final Test Accuracy and Test Loss scores.

2.15. Plot a normalized Confusion Matrix.

2.16. Write a script to extract and display the Top 3 False Positives or Top 3 False Negatives where the model was most confident in its wrong prediction. Display these images labeled with True Class vs. Predicted Class Probability.

2.17. Provide a brief error analysis explaining why the model made these confident mistakes on specific edge cases (e.g., visual overlap, background noise, or lighting variations. **(10 Marks)**

2.18. Predict class probabilities for an unseen test image and display the sample alongside a horizontal bar chart showing the final Softmax probability distribution.

2.19. Explain how raw model outputs (logits) are mathematically converted into probabilities using Softmax.

2.20. Discuss why neural networks can output overconfident probabilities for incorrect predictions and how this relates to model calibration. **(3 Marks)**

Oral Presentation (Mandatory) **(2 Marks)**

Deliver a 10-minute group presentation clearly communicating your model choice, training results, and key findings.

Submission Requirements

- **Report:** A single PDF document with the following sections:
  - Title Page (Project Title, Group Members' Names and IDs, Date).
  - Part 1: Research and Critical Analysis.
  - Part 2: Implementation and Evaluation.
  - References (**IEEE** style).
- **Code:** A single Jupyter Notebook file (.ipynb) containing all the Python code used for the implementation, with clear comments and explanations for each step.
- **Formatting:** Times New Roman, 12-point font, 1.5 line spacing, justified text.
- **Page Limit:** 10-35 pages.
- **File Naming:** Group_NO_Image_Recognition.pdf and Group_NO_Image_Recognition.ipynb, where NO is your group number.
- **Group Contributions:** Include a section in the report detailing the roles and contributions of each group member.

Group Assignment Rubrics

Part A: Situation/Scenario problem solving (10%)

Criteria	Weightage	Outstanding (9-10)%	Mastering (7-8)%	Developing (4-6)%	Beginning (1-3)%
Apply knowledge of Deep Neural Networks and related machine learning methods.	10%	Thorough documentation. Knowledge base is completely addressed the problem domain. Ability to stimulate conversation and engaging in conversation. Evidence of sound understanding. The solutions provide good recommendations. Extensive amount of critical insight and evaluation included.	Evidence of wider understanding. The answer of solution provides credible recommendations. Fairly well documented solution. A systematic approach to development and evaluation is used. Fair amount of critical insight and evaluation included.	Barely completed all answer. Minimum compliance to the domain problem. The solution is function, but discontinuity/missing observed in several instances. Solution provided does not engage in a full-fledge conversation. Knowledge and reasoning lacking in some instances.	Incomplete answer. The answer barely addressed the problem domain. little or no explanation of solution provided.

Part B: Mini Project (20%)

Criteria	Weightage	Outstanding (16-20)%	Mastering (12-15)%	Developing (8-11)%	Beginning (0-7)%
Apply knowledge of Deep Neural Networks and related machine learning methods.	20%	The answer is valid and acceptable. The program/code/evidence is correct. The justification/ description is well written.	The answer is valid and acceptable. The program/code/evidence is correct. But the justification/ description is not well written.	The answer is acceptable. The program/code/evidence has minor mistakes. But the justification/description is not well written.	The answer is not valid and acceptable. The program/code/evidence is missing or is not correct. The justification/description is not accurate and/or not clear and/or and not valid.

		Meets all written report formatting and submission requirements. The verbal presentation is, delivered confidently within the 10-minute limit, and clearly communicates the project's key aspects (Model, Results, and Insights).	Minor written report formatting issues, or one required document element is missing. The verbal presentation is complete and within time, but could be clearer in communicating one key aspect.	Several written report formatting/submission requirements are not met. The verbal presentation significantly exceeds/falls short of the time limit, or the content is difficult to follow, not showing proper preparation.	Major submission issues (e.g., wrong format, missing key report elements). The verbal presentation is not delivered or is irrelevant/incomplete, failing to convey the project's essence.